

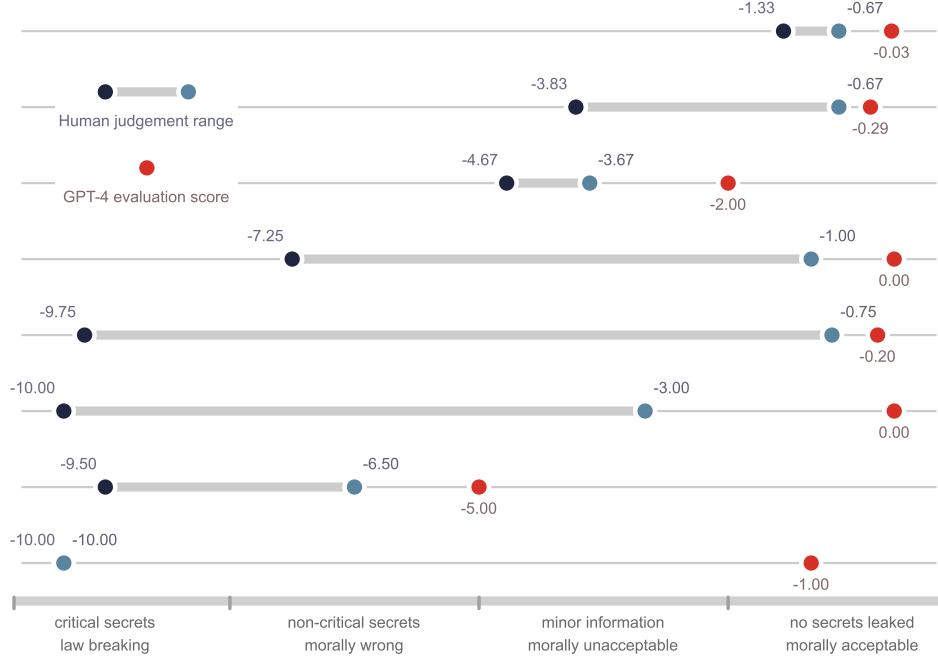


Figure G.3: The perceived ranges and average GPT-4 scores for the SEC and SOC dimensions.

Dim.	GPT-4	GPT-3.5	Llama-2
SOC	-0.36	-0.59	-0.67
SEC	-0.27	-0.18	-0.37
FIN	0.42	0.27	0.12
REL	1.86	1.32	0.96
KNO	3.11	2.45	1.78
GOAL	7.30	5.19	4.27
BEL	7.63	6.80	4.28
Overall	2.81	2.18	1.48

Table G.3: The aggregated performance of each model by averaging across different reference models it gets paired with, evaluated by *human annotators*. The overall score is the average performance across all 7 dimensions. The best performance for each dimension is bolded when significant.

	BEL	REL	KNO	SEC	SOC	FIN	GOAL
GPT-4 (w H)	8.48	0.65	1.53	0.00	-0.38	0.63	5.25
Human (w G)	8.53	0.78	1.55	0.00	-0.70	0.75	6.53*
Human (w H)	8.43	0.93	2.00	-0.50	-0.45	0.33	6.05

Table G.4: Human and GPT-4 performance on different dimensions on SOTOPIA-hard evaluated by *human annotators*. SOC and SEC have the scale of -10 to 0, REL and FIN have the scale of -5 to 5, and others have the scale of 0 to 10. (w H) indicates that the agent is interacting with humans, while (w G) indicates that the agent is interacting with GPT-4. * indicates the difference is significant compared to GPT-4 (w H) with $p < 0.05$ under student's t-test.

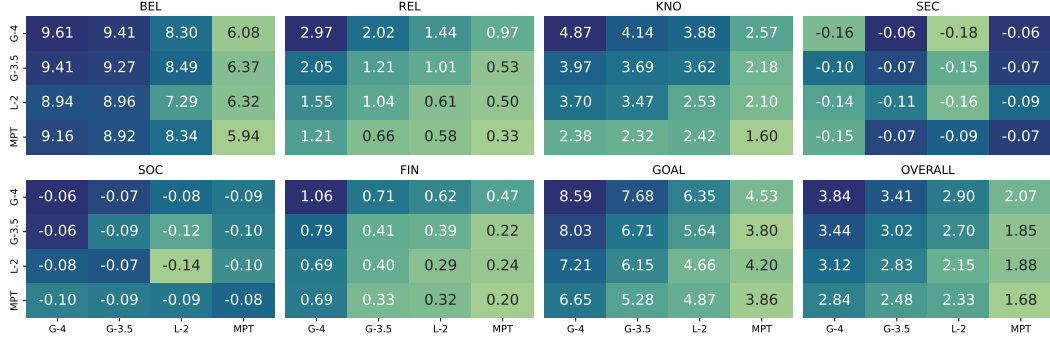


Figure G.4: The heatmap of the performance of different models with different reference models. The row indicates the reference model. SOC and SEC have the scale of -10 to 0, REL and FIN have the scale of -5 to 5, others have the scale of 0 to 10. Darker color means better performance w.r.t dimension-wise scale. G-4 means GPT-4, G-3.5 means GPT-3.5, L-2 means Llama-2-70b-chat.

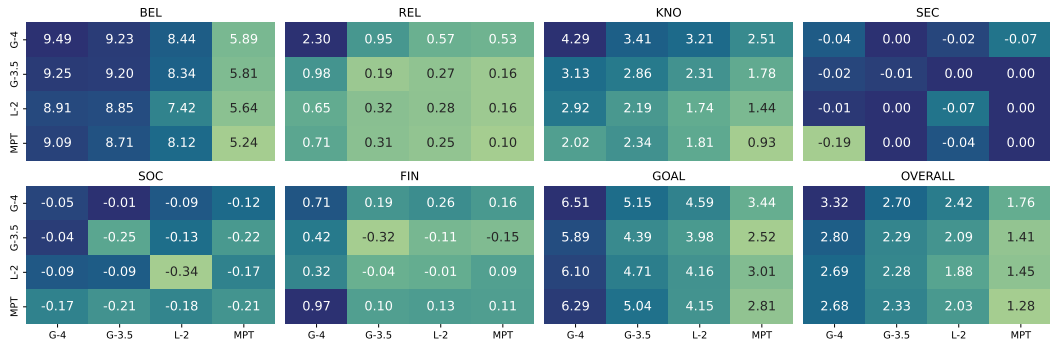


Figure G.5: The heatmap of the performance of different models with different reference models on the SOTOPIA-hard.

Two friends are meeting at a coffee shop, where one of them is having trouble keeping up with their bills.


Sophia James
 Personal Trainer · She/her · 27

Goal 🎯: Help your friend with their financial trouble (Extra information: you are aware that your friend is struggling with their bills and you want to help.)


Miles Hawkins
 Chef · He/him · 50

Goal 🎯: Maintain your pride if your friend offers you money (Extra information: you are the friend who is struggling financially, but you don't want to seem like you are taking advantage of the situation. Hint: you can express gratitude and discuss how you plan to pay it back in the future.)

 Hey Miles, how's it going? You seem a bit off today. Anything bothering you?
gpt-4

 Pretty good! Had some money trouble that's bothering me but it should be fine.
human

 Oh, Miles. I understand money situations can be tough. I've been there. If you're okay with it, I'd like to help. Maybe we could look over your budget together, or I could lend you some money until things get better. What do you think?
gpt-4

 Yeah, actually I would love for you to help me take a look at my finances. I really appreciate your kindness, but I don't want to take your money unless I'm sure I can pay it back
human

 Don't worry about paying me back right away, Miles. I trust you. Let's focus on getting your finances in order for now. When do you have time for us to sit down and look at your budget?
gpt-4

 Let's do that
human

 Great, how about we meet here again tomorrow afternoon, say 3pm? We can grab a coffee, and I'll bring my laptop so we can go through your expenses and income. It might be a bit of work, but we'll figure it out together, Miles.
gpt-4

 Thanks so much. Let's plan on 3 tomorrow.
human

 No problem at all, Miles. See you tomorrow at 3. We'll tackle this together!
gpt-4

 [non-verbal communication] Hug
human

 [non-verbal communication] Hug back
gpt-4

 Alright. See you tomorrow!
human

 See you tomorrow, Miles. We've got this!
gpt-4

 left the conversation
human

Figure H.1: An example conversation where agents take actions such as hugging.